AI MODELS & LOCAL INFERENCE

Why Multi-Turn Agents Burn Compute

Why multi-turn LLM agents waste 70% of compute re-evaluating system prompts and how Radix Trees fix it.

THE BOTTLENECK

In a 15-step agent loop, the system prompt, tool schemas, and prior trajectory steps are sent on every single turn. Standard serving engines discard or naively recompute KV cache states across branching tool calls.

```
70% of prefill FLOPs wasted re-reading static tool definitions
```

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github.com/sgl-project/sglang →

KV Cache as a Radix Tree

Architecture Shift

ARCHITECTURE SHIFT

Instead of treating prompts as isolated requests, SGLang maintains a Least-Recently-Used (LRU) Radix Tree of token prefixes in GPU VRAM. Shared prefixes across subagents point to the exact same KV memory blocks.

```
RadixTree: [System Prompt] → [Tool Schemas] → [Branch A / Branch B]
```

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Up to 5x Higher Agent Throughput

Benchmark Evidence

BENCHMARK EVIDENCE

On multi-turn agent trajectories, tree-of-thought searches, and few-shot RAG pipelines, RadixAttention cuts Time-To-First-Token (TTFT) by 60–80% while multiplying concurrent batch throughput.

```
Throughput: up to 5.0x vs baseline without prefix tree caching
```

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github.com/sgl-project/sglang →

VRAM Eviction & Cache Thrashing

The Engineering Catch

THE ENGINEERING CATCH

Radix trees only help if your prompts share deterministic prefixes. If you inject dynamic timestamps or random UUIDs at the top of your system prompt, you invalidate the entire prefix tree on line 1.

```
Rule: Always put static system prompts & tools first; dynamic state last
```

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Drop-In OpenAI-Compatible Runtime

Takeaway & Sandbox

TAKEAWAY & SANDBOX

SGLang ships under an Apache 2.0 license and exposes a standard OpenAI-compatible HTTP endpoint. Always benchmark on your actual prompt prefix length inside an isolated container before migrating.

```
Repo: github.com/sgl-project/sglang (Apache 2.0)
```

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